Question:

A 66-year-old farmer is being evaluated for abnormal lung findings on a low dose chest CT scan obtained as part of his lung cancer screening. He has a 50-pack-year smoking history and has been hesitant to quit. He has a non-productive cough but brushes it away saying he is not bothered by it. He denies ever coughing up blood, breathlessness, chest pain, fatigue, or weight loss. He has never sought any medical care and states that he has always been in good shape. He consumes alcohol moderately and uses marijuana occasionally. He lives with his wife and has not traveled recently. On physical examination, his temperature is 37.1°C (98.8°F), blood pressure is 148/70 mm Hg, and pulse rate is 95/min. His BMI is 32 kg/m2. A general physical examination is unremarkable. Coarse breath sounds are present bilaterally. The cardiac exam is normal. Laboratory studies show a normal complete blood count and comprehensive metabolic panel. A follow-up high-resolution CT scan is performed that shows small irregular subcentimeter pulmonary nodules, several of which are cavitated in both lungs, predominantly distributed in the upper and middle zones. There is no mediastinal or hilar lymphadenopathy. A transbronchial needle aspiration of the lesion is performed which shows a nodular pattern of abundant, granular, mildly eosinophilic cells with grooved nuclei with indented nuclear membranes and a chronic inflammation that consists primarily of eosinophils. Immunohistochemical staining reveals numerous cells that stain positive for S100 and CD1a. Which of the cells of the human immune system are responsible for this lesion?

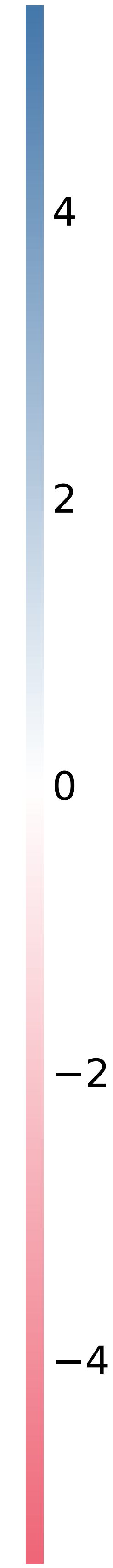
Answer Choices:

A. T lymphocytes

B. Natural killer cells

C. Dendritic cells

D. Ciliary epithelium



Reasoning + Answer:

<think> The patient had a low dose chest CT scan for lung cancer screening. The findings included subcentimeter pulmonary nodules with cavitation. The biopsy and immunohistochemical staining were discussed. T lymphocytes, natural killer cells, dendritic cells, and ciliary epithelium were possible immune cell types. </think><answer>Dendritic cells</answer>